

2. Microservices with API gateway

Creating Microservices for account and loan

Account Microservice

- Create folder with employee id in D: drive
- Create folder named 'microservices' in the new folder created in previous step. This folder will contain all the sample projects that we will create for learning microservices.
- Open <https://start.spring.io/> in browser
- Enter form field values as specified below:
 - **Group:** com.cognizant
 - **Artifact:** account
- Select the following modules
 - Developer Tools > Spring Boot DevTools
 - Web > Spring Web
- Click generate and download the zip file
- Extract 'account' folder from the zip and place this folder in the 'microservices' folder created earlier
- Open command prompt in account folder and build using mvn clean package command
- Import this project in Eclipse and implement a controller method for getting account details based on account number. Refer specification below:
 - Method: GET
 - Endpoint: /accounts/{number}
 - Sample Response. Just a dummy response without any backend connectivity.

```
{ number: "00987987973432", type: "savings", balance: 234343 }
```

 **spring initializr**

Project

☐ Gradle - Groovy

☐ Gradle - Kotlin

☒ Java

☐ Kotlin

☐ Groovy

☒ Maven

Spring Boot

☐ 4.0.0 (SNAPSHOT)

☐ 3.5.4 (SNAPSHOT)

☒ 3.5.3

☐ 3.4.8 (SNAPSHOT)

☐ 3.4.7

Project Metadata

Group

com.cognizant

Artifact

account

Name

account

Description

Demo project for Spring Boot

Package name

com.cognizant.account

Packaging

☒ Jar

☐ War

Java

☐ 24

☐ 21

☒ 17

Dependencies

ADD DEPENDENCIES... CTRL + B

Spring Boot DevTools

DEVELOPER TOOLS

Provides fast application restarts, LiveReload, and configurations for enhanced development experience.

Spring Web

WEB

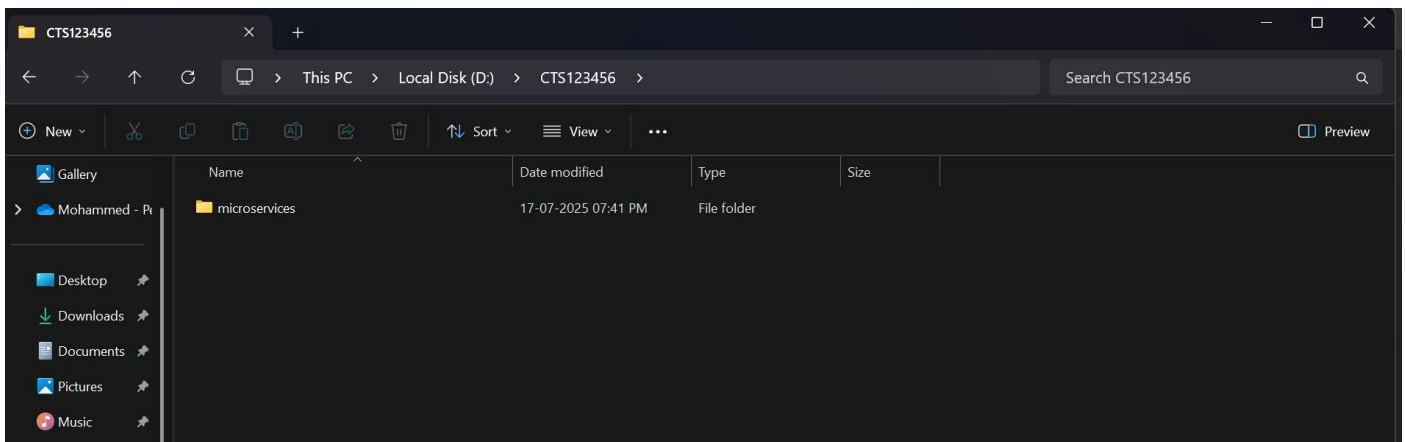
Build web, including RESTful, applications using Spring MVC. Uses Apache Tomcat as the default embedded container.

GENERATE CTRL + G

EXPLORE CTRL + SPACE

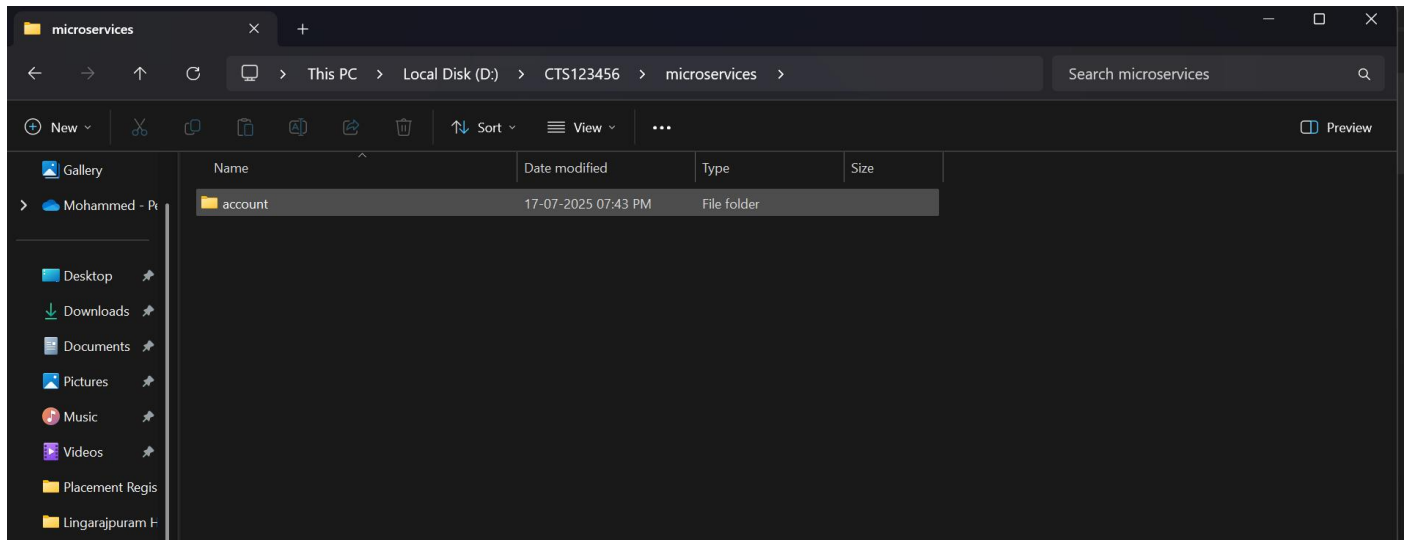
...

Create a folder in D drive

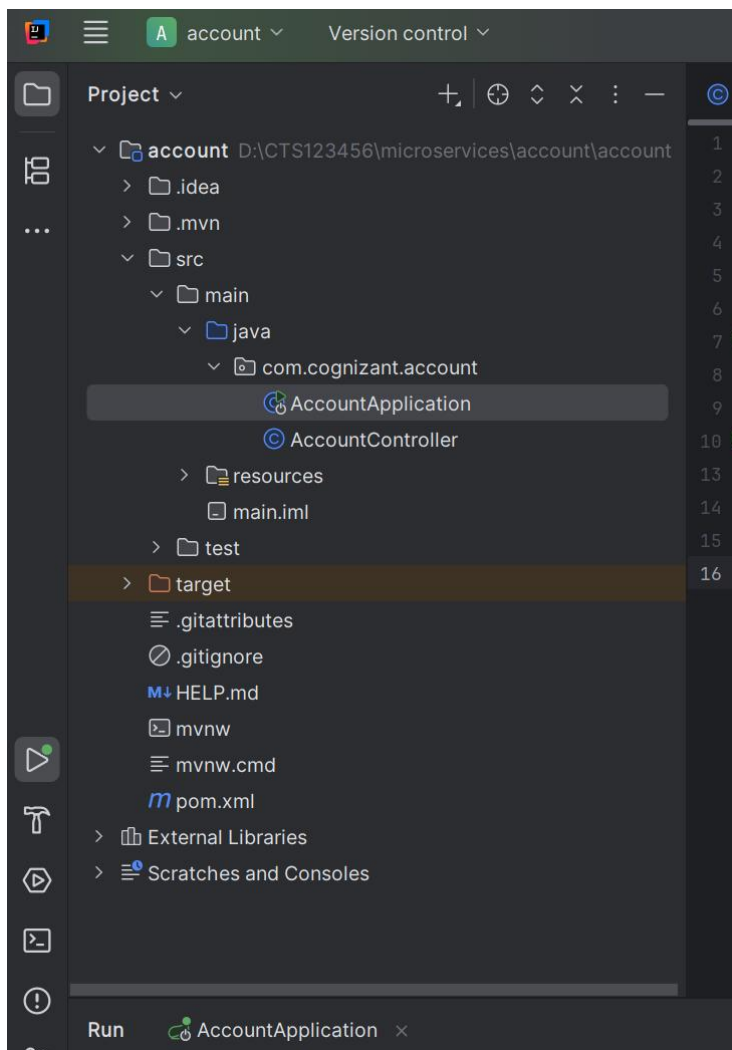


Create a folder named microservices inside the created folder

Then, extract the spring initializr file and paste it inside the “microservices” folder



Folder Structure:



AccountController.java

```
package com.cognizant.account;

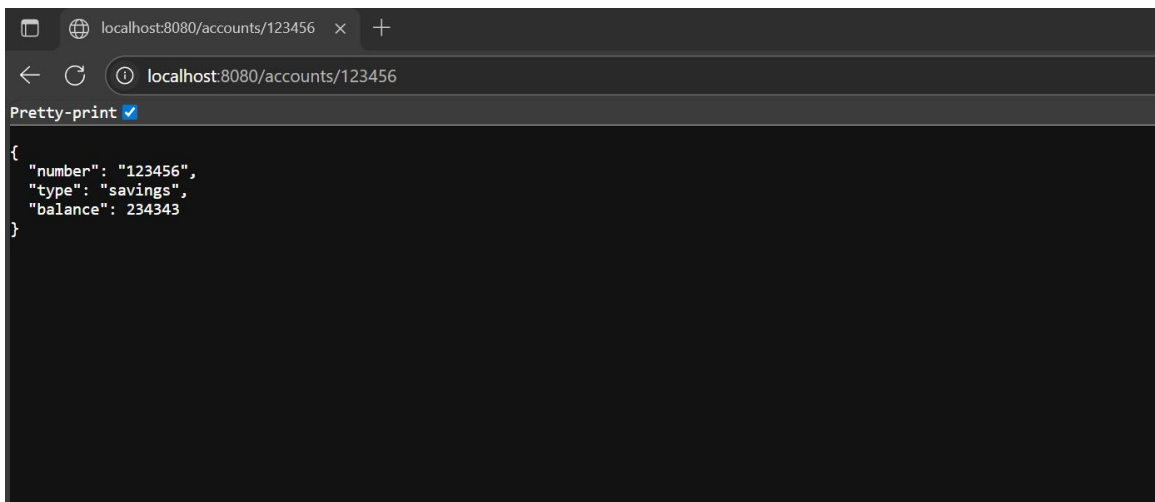
import org.springframework.web.bind.annotation.*;

@RestController
@RequestMapping("/accounts")
public class AccountController {

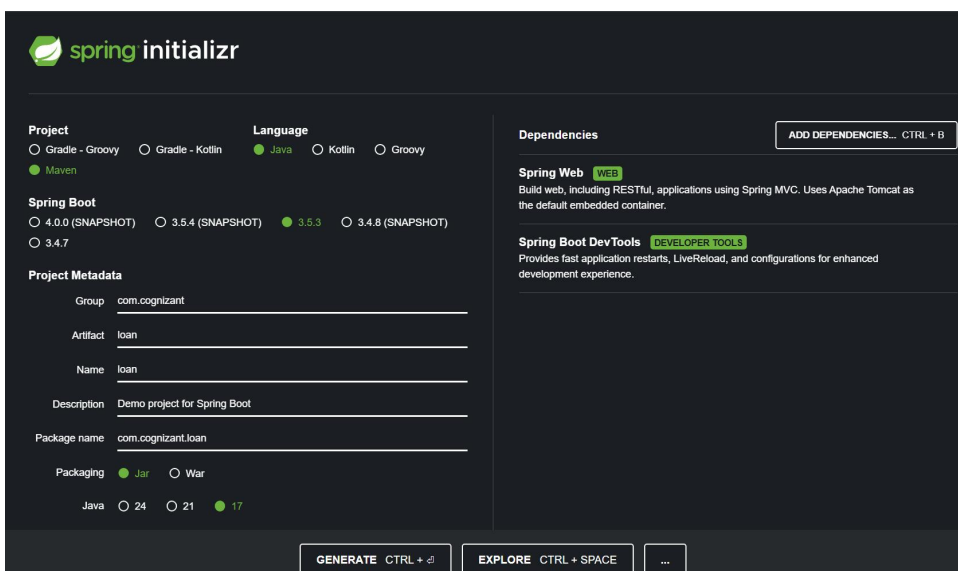
    @GetMapping("/{number}")
    public Account getAccount(@PathVariable String number) {
        return new Account(number, "savings", 234343);
    }

    record Account(String number, String type, double balance) {}
}
```

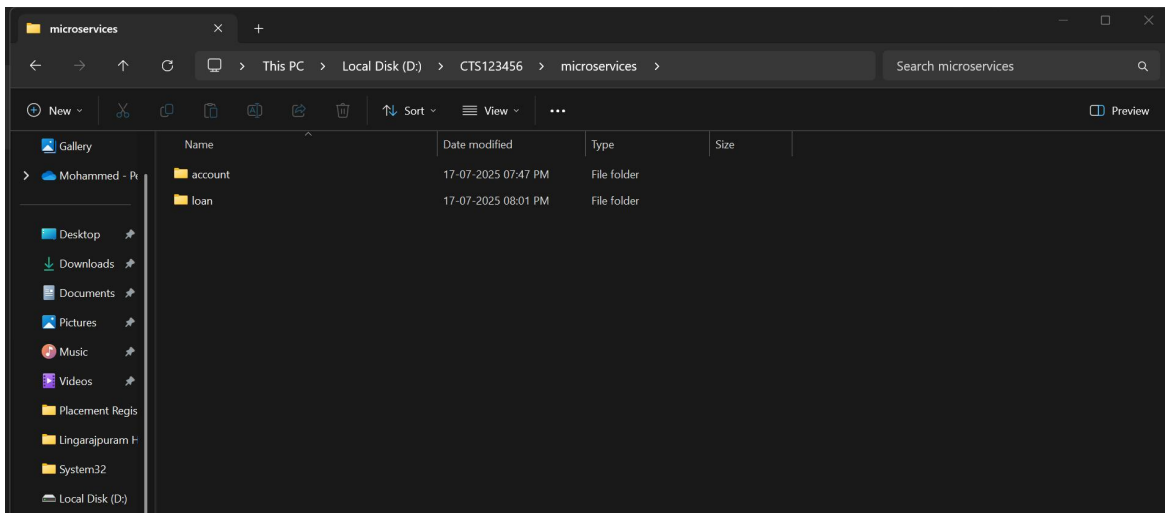
Then, run AccountApplication.java and go to localhost:8080/accounts/123456



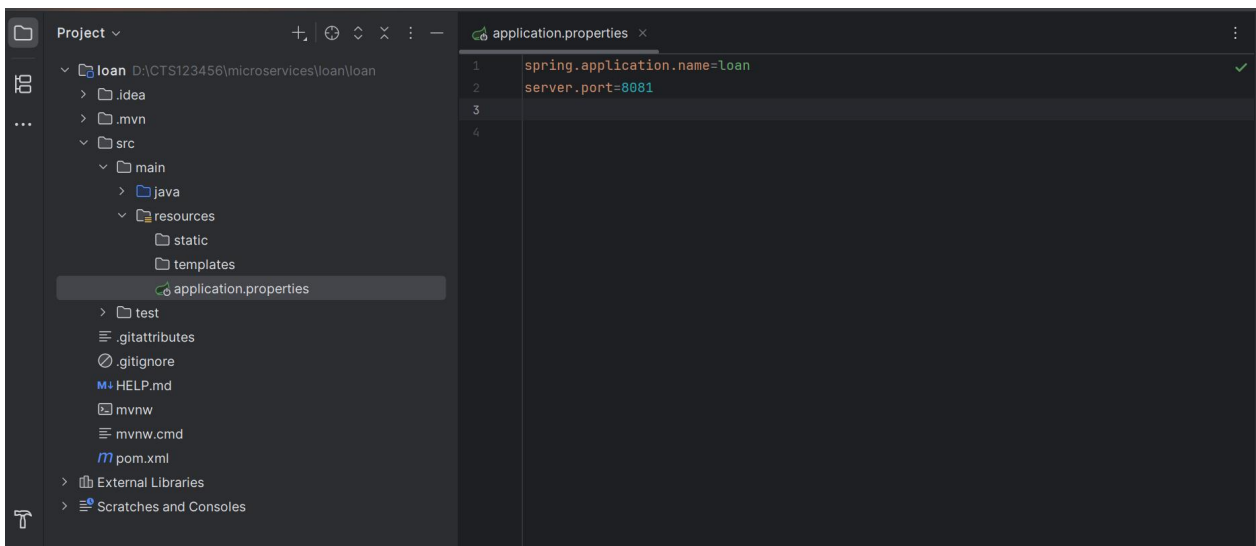
Creating Loan microservice:



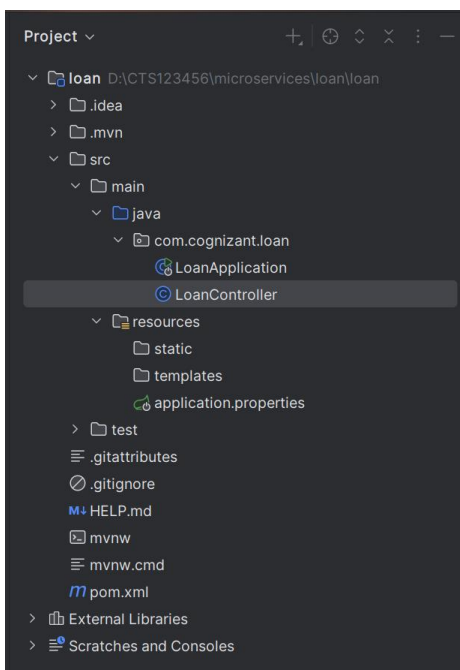
Then move it to the same old folder,



Set custom port for the loans project by adding → `server.port=8081`



Folder structure:



LoanController.java

```
package com.cognizant.loan;

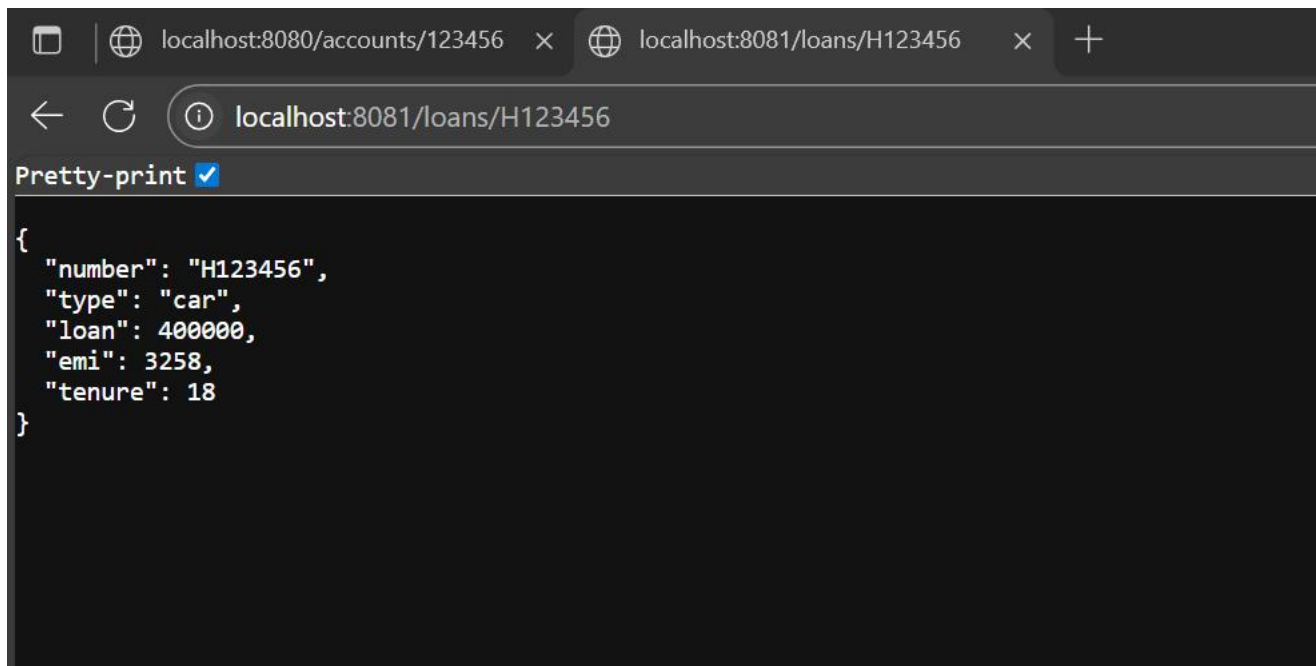
import org.springframework.web.bind.annotation.*;

@RestController
@RequestMapping("/loans")
public class LoanController {

    @GetMapping("/{number}")
    public Loan getLoan(@PathVariable String number) {
        return new Loan(number, "car", 400000, 3258, 18);
    }

    record Loan(String number, String type, double loan, double emi, int tenure) {}
}
```

Run LoanApplication.java and go to <http://localhost:8081/loans/H123456>



I now have two microservices,

- 1) Account on port 8080
- 2) Loan on port 8081.